

22. (2.5 points)

A start-up insurance company writes exclusively private passenger auto insurance, offering both liability and physical damage coverages. Based on the latest Enterprise Risk Management (ERM) review, management is confident in its capital utilization and plans to grow.

To attract and retain customers, the insurer is considering expanding into one of the following three lines of business:

- Renters
- Homeowners
- Livery conveyance (commercial coverage for insureds that use their personal vehicle and drive part-time for ridesharing companies such as Uber, Lyft, etc.)

a. (0.25 point)

Explain why parameter risk is a key source of uncertainty in enterprise risk modeling.

b. (1.5 points)

Identify and briefly describe an aspect of parameter risk introduced by incorporating each of the three lines of business into the insurer's existing ERM model. Use a different type of parameter risk for each line of business.

c. (0.5 point)

Explain why parameter risk may be more significant for large insurers than small insurers.

d. (0.25 point)

Briefly describe how parameter risk can be reduced.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 22	
TOTAL POINT VALUE: 2.5	LEARNING OBJECTIVE(S): A1, A6
SAMPLE ANSWERS	
Part a: 0.25 point	
<u>Sample 1</u> Parameter risk is the random uncertainty in the frequency and severity distributions of each line of business since the parameters and forms of these distributions are only estimates. This is a significant risk for insurance of all sizes because it is not able to be diversified. <u>Sample 2</u> Parameter risk reflects the imperfection of the model itself. The sources of uncertainty in ERM come from process risk and parameter risk. Parameter risk, unlike process risk, can't be diversified away. This makes parameter risk a key source of uncertainty. <u>Sample 3</u> Parameter risk is a key source of uncertainty because if an estimated parameter does not accurately reflect the true underlying parameter, then the projected results could be highly underestimated or highly overestimated.	
Part b: 1.5 points	
<u>Sample 1</u> Renters: can introduce estimation risk. Estimating the form and parameters of a loss distribution requires data. This insurer doesn't have as this is a new line of business. Homeowners: can introduce projection risk. Homeowner is an even more difficult case. In addition to the Estimation risk cited with Renters, Homeowners is subject to even more uncertainty over time due to changes in macroeconomic conditions and extreme weather events. Livery: can introduce event risk, as it applies to ridesharing drivers, is still in its regulatory, legislative, and judicial infancy. A major court decision or legislative act could change the intent of policy coverage. <u>Sample 2</u> Renters: can introduce model risk. AS this is a new line of business the insurer may not be able to identify the appropriate model underlying the risk. Homeowners: can introduce systemic risk. Systemic risk operates on large number of individual policies. They are, therefore, non-diversifying. As homeowners value is highly impacted by inflation, systemic risk will be an issue for the insurer. Livery: can introduce estimation risk. Risk that since the insurer is expanding into this line of business, their data is limited and prone to bias and error. This will lead to estimated parameters being incorrect and biased if estimated from this data.	
Part c: 0.5 point	
Large insurance companies lower risk through diversification/volume. Parameter risk exists for each business segment and cannot be reduced by diversification/volume. With other risks	

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

mitigated through diversification/volume parameter risk make up a relatively larger risk for the large company.

Part d: 0.25 point

Sample Responses include any of the following:

- Having more data and better data quality
- More advanced regression procedures
- Testing sensitivity of results to the model (use multiple models to test)
- Combine /Weight together different Models
- Use Expert Judgment
- Test validity of results against available data.
- Use Compound simulation where parameters are selected from one distribution to be modeled in another
- Reduce any over-parameterization

EXAMINER'S REPORT

Candidates were expected to understand why parameter risk is an important piece of the total risk, which includes process risk and parameter risk. They were also expected to be able to identify and describe aspects of parameter risk.

Part a

Candidates were expected to show that they understood what parameter risk is and be able to relate it to **why** it is a key source of uncertainty (i.e. not merely provide a definition of parameter risk).

Common errors include:

- Describing parameter risk with no link to uncertainty of models.
- Saying that it is hard to estimate parameters.
- Just stating the aspects of parameter risk or defining parameter risk.

Part b

Candidates were expected to be able to identify three different aspects of parameter risk and be able to describe them in the context of entering into three new lines of business.

The following aspects of parameter risks were accepted:

- a. Estimation risk
- b. Projection risk
- c. Event risk
- d. Systematic risk
- e. Model risk

Common errors include:

- Using catastrophes as part of an event risk since a catastrophic occurrence does not necessarily mean your estimated parameters were wrong.
- Some candidates identified data error and specification error as aspects of parameter risk

Part c

Candidates were expected to understand why parameter risk for large insurers is more significant than in smaller insurers.

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

Common errors include:

- Stating that large insurers deal with more lines of business and therefore have more parameters to estimate.
- Stating that large insurers have more correlations amongst the business they write.

Part d

Candidates were expected to be able to identify at least one way to reduce parameter risk.
Candidates generally did well on this portion of the question.